



Numerical simulation of the molten pool in laser spiral spot welding of Al/Fe dissimilar metals

Shuang Huang^{a,b}, Han Yu^a, Ming Lou^{a,*}, Yongbing Li^{a,**}

^a Shanghai Key Laboratory of Digital Manufacture for Thin-walled Structures, Shanghai Jiao Tong University, Shanghai, China

^b State Key Laboratory of Long-Life High Temperature Materials, Dongfang Turbine Co., Ltd, Dongfang Electric Corporation, Deyang 618000, China

ARTICLE INFO

Keywords:

Simulation

Laser spiral spot welding

Al/Fe dissimilar metal

Molten pool

ABSTRACT

Laser spiral spot welding (LSSW), with precisely adjustable energy distribution, effectively increases bonding interface density and controls intermetallic compounds (IMCs) formation, serving as a key method for welding Al/Fe dissimilar structures. In this study, a three-dimensional multi-phase transient numerical model of Al/Fe LSSW was developed to elucidate the temperature evolution process and the molten pool flow mechanism during welding. The results demonstrate that the molten pool maintains an inverted triangular pyramidal morphology with curved edges, and its radius expands along the scanning path. The enhanced Marangoni stress in the outer region constrains the molten pool radius to be smaller than the scanning path radius. The flow velocity is higher in the inner ring, with a peak value near the keyhole wall. Peclet numbers are predominantly between 2 and 41 indicating that the convective heat transfer dominates the thermal transport within the molten pool. Furthermore, the molten pool of Fe, driven by the evaporation recoil pressure, impinges on the Al layer, causing partial melting and extrusion of Al into a U-shaped liquid film, which is the key mechanism for the formation of the Al-rich U-shaped solute band. In addition, thermal history analysis provides valuable insights into the solidification characteristics of the joint.

1. Introduction

In the development of lightweight thin-walled structures for new energy vehicles, joining dissimilar metal sheets has become a key trend, balancing structural strength and stiffness with the cost-effective demands of lightweight design [1–3]. However, the substantial disparities in the physical and chemical properties between Al and Fe present significant challenges for attaining a reliable joint. Defects such as cracks and pores are prone to form in the welded region, which significantly undermine the performance of the joint [4–6]. Laser welding is widely considered a viable solution for joining dissimilar metals, owing to its advantages of high energy density, a narrow heat-affected zone and minimal welding-induced distortion [7–9]. Specifically, laser spiral spot welding (LSSW) enables reliable joining by offering good adaptability to gap variations, high density of bonding interfaces, and reduced pitch between weld spots.

Deng et al. [10] employed LSSW on overlapping galvanized Fe sheets and observed that welding heat accumulates progressively as the scanning path lengthens. It recommended that the linear energy input during

the later stages of welding be substantially lower than that in the initial phase to prevent overheating of the weld seam. Xiong et al. [11] investigated the relationship between the porosity and the process parameters in LSSW. The results show that reducing the spiral scanning spacing prolongs the molten pool lifetime and enhances the keyhole stirring effect, thereby facilitating the escape of the pores. Li et al. [12] examined the effect of spiral scanning spacing on the microstructure of Cu/Al dissimilar material joints. With the increase of spiral spacing, the mutual diffusion of Al and Cu was weakened, resulting in an interfacial morphology characterized by a wavy structure with continuous blocky θ -CuAl₂ phases. Wang et al. [13] compared the effects of different scanning paths (spiral and polygonal) on the microstructure and mechanical properties of Al/Cu joints by nanosecond laser. The experimental results show that although the microstructure and joint strength are similar under both paths, the processing time for the spiral path (0.3 s) is significantly shorter than that for the polygonal path (0.6 s), corresponding to a 100% improvement in welding efficiency. These researches demonstrate that LSSW can effectively improve the mechanical properties of joints compared with traditional laser welding.

* Corresponding author.

** Corresponding author.

E-mail addresses: louming@sjtu.edu.cn (M. Lou), yongbinglee@sjtu.edu.cn (Y. Li).

<https://doi.org/10.1016/j.jmrt.2026.05.092>

Received 8 April 2026; Received in revised form 5 May 2026; Accepted 8 May 2026

Available online 10 May 2026

2238-7854/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Given the extremely high heating and cooling rates inherent in the process, directly measuring the molten pool flow and obtaining precise temperature distributions during LSSW of dissimilar metals poses substantial difficulties. Indhu et al. [14] and Pan et al. [15] used COMSOL software to establish a two-dimensional transient model of Al/Fe laser welding to investigate the temperature distribution, heating and cooling time during welding under different laser energy density. The process of temperature distribution of interface was discussed in detail. Xie et al. [16] used ANSYS Fluent to establish a three-dimensional transient model of laser lap welding of Al/Fe dissimilar metals. It was found that the flow of molten pool was mainly driven by surface tension, and convection and diffusion mechanisms of metal mass transfer in molten pool. Currently, researches on Al/Fe welding models primarily focus on the influence of process parameters on the molten pool during linear welding. Under the spiral path, the mass transfer between Al/Fe and the flow behavior of molten pool are still relatively scarce.

In this study, the physical mechanism of the metal mixing during Al/Fe lap laser welding with a spiral path was investigated by both experimental and simulation approaches. A computational fluid dynamics (CFD) model was developed to reveal the intrinsic physical mechanism of the metal mixing process by predicting heat transfer behavior, fluid flow pattern and metal mixing characteristics within the molten pool. Then, the spatial distributions of temperature, driving forces, fluid flow, and metal mixing in the molten pool were analyzed systematically. The temperature field and flow field characteristics of the molten pool were revealed during LSSW of Al/Fe dissimilar metals, which is of great significance for further understanding the weld formation mechanisms and promoting the optimization of welding process.

2. Mathematical model and numerical simulation procedure

The calculation domain of the model is set as an air-Fe-Al three-layer structure with a length $\times$ width $\times$ height of 10 mm $\times$ 10 mm $\times$ 3.5 mm, where the thickness of the air layer, high-strength Fe and aluminum alloy are 1.0 mm, 1.0 mm and 1.5 mm respectively. The solution domain and boundary conditions are shown in Fig. 1. In order to further improve the calculation speed, the grid refinement is only carried out in the areas such as the molten pool and the heat affected zone. The minimum size of the grid is set to 0.01 mm $\times$ 0.01 mm $\times$ 0.01 mm, and the maximum size of the grid in other areas is 0.1 mm $\times$ 0.1 mm $\times$ 0.1 mm. The total number of cells is 594600, and the number of nodes is 561561.

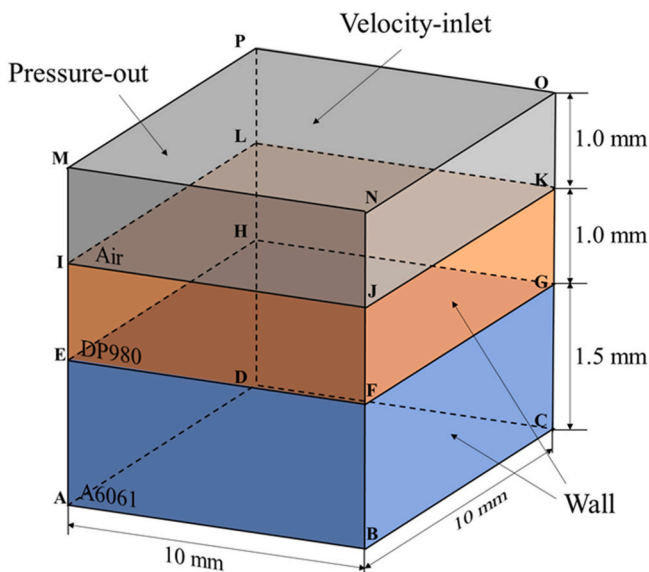


Fig. 1. Schematic diagram of boundary conditions.

2.1. Laser heat source

To account for the gradual decay of heat flux along the thickness direction during laser deep penetration welding, a Gaussian rotating volumetric heat source is employed to simulate the thermal effects of the laser within the molten pool. The expression is as follows [17]:

$$q(r, z) = \frac{9Q}{\pi R_0^2} \frac{e^3 - 1}{e^3} \exp\left(-\frac{9r^2}{\ln\left(\frac{H}{z}\right) R_0^2}\right) \quad (1)$$

2.2. Tracking of phase boundary

The volume of fluid (VOF) method is used to track the free surface of the phase boundary. F is defined as the volume fraction of the fluid in the grid. The continuity equation is as follows:

$$\frac{\partial F}{\partial t} + (\mathbf{V} \cdot \nabla) F = 0 \quad (2)$$

$$\sum_i F = 1 \quad (3)$$

The i represents different phases, where $i = 1$ is the air phase, $i = 2$ is the Al phase, and $i = 3$ is the Fe phase. The region with a fluid volume fraction of $0 < F < 1$ in the grid is the free surface of the phase boundary. In the post-processing, $F = 0.5$ was selected as the interface between air-Fe-Al. The volume fraction equation is calculated as explicit, and the minimum volume fraction (Volume Fraction Cutoff) is set to 10^{-6} . When the volume fraction is less than the set value, the volume fraction of the phase is determined to be 0. In order to improve the convergence of the model, the Courant Number is set to 0.25.

2.3. Boundary conditions

The top surface (MNOP) in the computational domain is set as the velocity inlet, and the gas phase side (MNJI, NOLJ, POKL and MPLI) is set as the pressure outlet. The side of the Al and Fe in the calculation domain of the workpiece is set to the wall boundary, mainly considering thermal convection and thermal radiation. The wall boundary conditions are described as follows [18]:

$$-k \frac{\partial T}{\partial \vec{n}} = -h_A(T - T_0) - \sigma \varepsilon(T^4 - T_0^4) \quad (4)$$

where $\vec{n}$ is the surface unit normal vector, h_A is the convective heat transfer coefficient, σ is the Stefan Boltzmann constant, ε is the radiation emissivity, and T_0 is the ambient temperature.

Due to the gas/liquid free surface (IJKL), the energy balance mainly considers the heat loss caused by thermal convection, thermal radiation and surface evaporation. The thermal boundary conditions can be expressed as:

$$-k \frac{\partial T}{\partial \vec{n}} = q - h_A(T - T_0) - \sigma \varepsilon(T^4 - T_0^4) - q_{evap} \quad (5)$$

where q is the laser heat flux, and q_{evap} is the heat loss caused by the evaporation of liquid metal. Ignoring the energy loss caused by the volume expansion of metal vapor, the equation can be expressed as:

$$q_{evap} = H_{lv} \frac{\partial(m_{lv} - m_{vl})}{\partial A} \quad (6)$$

where H_{lv} is the heat transfer during the phase transition from liquid phase to gas phase, A is the mass exchange area, m_{lv} is the mass conversion rate from liquid phase to vapor phase, and m_{vl} is the mass conversion rate from vapor phase to liquid phase. The thermophysical parameters of Fe and Al are listed in Table 1.

In the calculation setup, the second-order upwind scheme is used for

Table 1

Thermophysical property parameters of DP980 and A6061.

Thermophysical parameters	DP980	A6061
Solidus temperature (K)	1680	860
Liquidus temperature (K)	1727	900
Evaporation temperature (K)	3100	2700
Density (kg/m ³)	7800	2660
Dynamic viscosity (kg/m•s)	0.006	0.0012
Specific heat (J/kg•K)	475	1198
Latent heat of fusion (J/kg)	2.47×10^5	3.87×10^5
Latent heat of evaporation (J/kg)	6.34×10^6	1.05×10^7
Surface tension coefficient (N/m)	1.2	0.781
Temperature coefficient of surface tension (N/m•K)	-4.9×10^{-4}	-1.55×10^{-4}
Radiation emissivity	0.4	0.3

the spatial discretization of the momentum and energy conservation equations, and the first-order upwind scheme is used for the spatial discretization of the laminar kinetic energy and laminar dissipation rate. For the pressure-velocity coupling method, the SIMPLE scheme is adopted, and the Geo-Reconstruct method is used for the spatial discretization of the volume fraction. The calculations are performed on a high-performance workstation (model: DESKTOP-G5MOPF6) using 32 AMD CPU cores, with a time step size set to 1×10^{-5} s.

3. Experimental procedure

The schematic diagram of LSSW of Al/Fe dissimilar metals is depicted in Fig. 2. The spiral spot oscillation trajectory is employed in this study, and the laser beam scanning in an inward-to-outward helical pattern. The welding equipment is a Trumpf TruDisk 8000 disc laser, which possesses the maximum output power of 8000 W, the minimum fiber core diameter of 50 μ m, and the laser wavelength is 1030 nm. Before welding, the plate was polished and cleaned with sandpaper and acetone solution respectively. During welding, the laser power was 1150 W, the welding speed was 170 mm/s, and the defocusing distance was -3 mm. Additionally, the 99.99 % nitrogen gas was used in the welding process, with a flow rate of 20 L/min.

4. Results and discussion

4.1. Validation of simulation model

To verify the reliability of the model, the simulated weld pool geometries were compared with experimental results. Fig. 3 (a) and 3 (b) respectively depict the upper surface morphology of the inner and outer welding layers. The experimental molten pools between the inner and outer layers exhibit similar dimensions, with an average length of 1.14 mm and a width of 0.57 mm. In the simulation results, the contour of the molten pool was delineated with the liquidus temperature of DP980 Fe serving as the upper limit and the solidus temperature as the lower limit. The corresponding simulation outcomes are presented in Fig. 3 (c) and 3 (d), indicating an average pool length of 1.09 mm and width of 0.58 mm. The close correspondence between the simulated molten pool profiles and the actual weld geometry confirms the validity of the numerical model.

Fig. 4 shows the cross-section of the weld profile for the experimental and the corresponding simulation results. The weld penetration and weld width of the four passes are measured, as shown in Table 2. It can be seen from the comparison that the simulated molten pool and the actual weld can be in good agreement in the depth direction, but the upper surface melting width of the simulation result is slightly smaller than the actual weld profile, as shown in Fig. 4. This is because the liquid metal flowing backward from the keyhole in the experimental welding process will continue to broaden the molten pool. Through the above comparison, the error of the simulation model is less than 15.8 %, which can effectively simulate the LSSW process of Al/Fe.

4.2. Molten pool characteristics on the upper surface of the Fe side

Fig. 5 illustrates the temperature field distribution on the upper surface of the molten pool and the three-dimensional morphology of the Fe layer (blue layer) during the inner ring welding process at different time intervals. At 0.030 s (Fig. 5(a)), the laser beam irradiates the left side of the specimen, and the upper surface of the molten pool exhibits a crescent-shaped profile consistent with the welding scanning path. The weld width reaches its maximum at the keyhole and gradually decreases toward the trailing edge. Additionally, as depicted in Fig. 5(b), the molten pool maintains a similar crescent-shaped contour at various

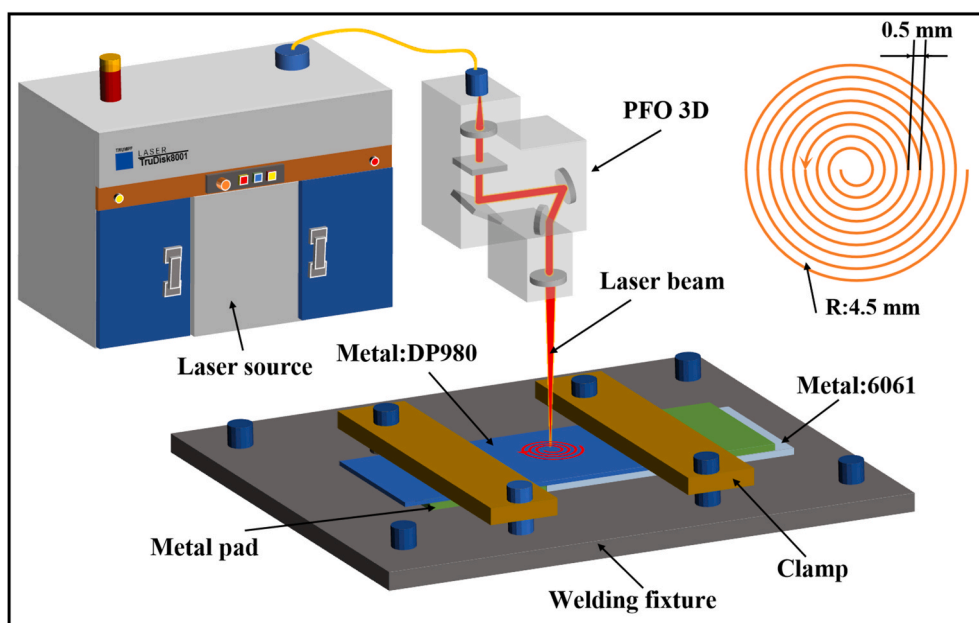


Fig. 2. Schematic of laser spiral fusion welding setup.

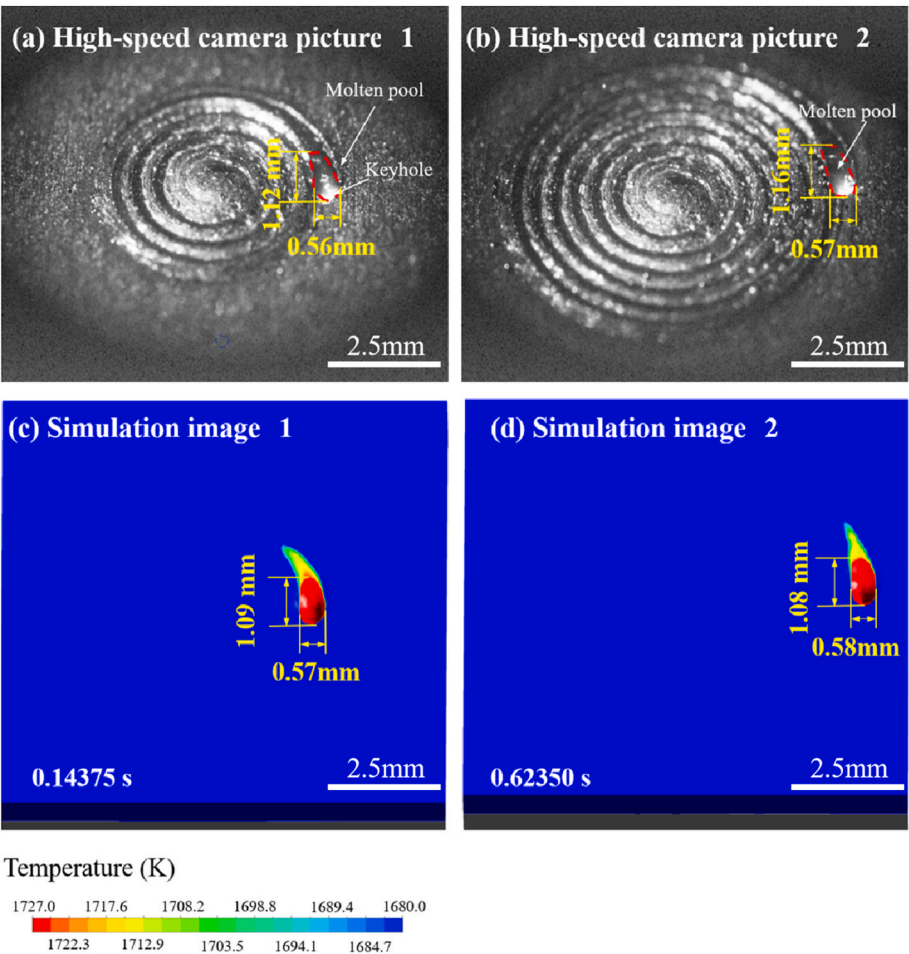


Fig. 3. Comparison of experimental results and numerical simulation results: High-speed camera morphology of (a) inner ring and (b) outer ring, numerical results of (c) inner ring and (d) outer ring.

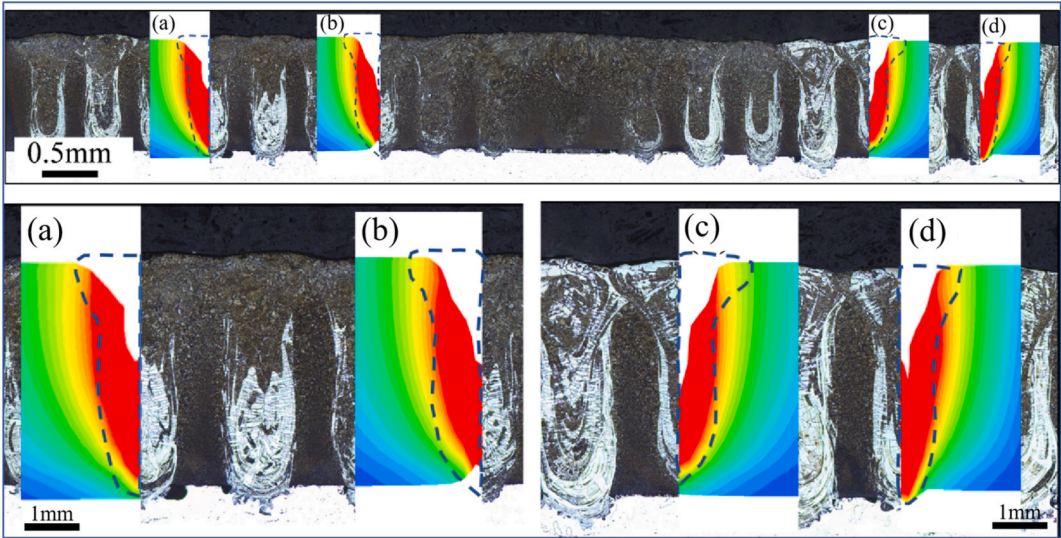


Fig. 4. Comparison of actual and numerically simulated profiles of weld cross sections.

depths. The maximum depth of the molten pool is observed at the keyhole, gradually diminishing toward the tail region, thereby forming an inverted triangular cone with a distinct angle.

As illustrated in Fig. 5(b)–(h), irregular and intermittently distributed concave features appear on the upper surface of the Al layer (gray

layer). The foremost region of these concavities is located directly beneath the keyhole, corresponding to the deepest part of the weld. This indicates that the concavities in the Al layer are formed by the impingement of the Fe molten pool into the underlying Al layer, leading to localized melting of the Al and subsequent displacement of the molten

Table 2
Weld dimension verification.

No.	Actual penetration (mm)	Simulation penetration (mm)	Error (%)	Actual middle part Width (mm)	Simulation middle part Width (mm)	Error (%)
1	1.07	1.04	2.8	0.19	0.22	15.8
2	1.08	0.95	12.0	0.21	0.21	0.0
3	0.93	0.91	2.2	0.17	0.17	0.0
4	1.04	1.04	0.0	0.17	0.15	11.8

Al through extrusion. The intermittent and non-uniform distribution of these concavities, along with their varying depths, suggests that non-periodic fluctuations in the depth of the Fe molten pool persist throughout the welding process.

Compared with the inner ring weld, the outer molten pool (Fig. 6 (a) (c) (e) and (g)) radius decreases along with the reduction in the welding

scanning path, and the temperature field distribution around the upper surface of the weld pool becomes more dispersed. Meanwhile, the shape of the molten pool in Fig. 6 (b) (d) (f) and (h) retains a curved inverted triangular cone shape, which is similar to the shape of the inner ring molten pool. The different Al–Fe mixed zones (the concave regions of the Al layer) are still discontinuously distributed, but their continuity is improved. This indicates that the stability of the Fe molten pool is enhanced at the outer-pass welding.

Fig. 7 shows the variations in molten pool morphology under different welding orientations for the inner and outer rings. The red areas represent the liquid metal pool with a temperature higher than the liquidus, while the regions between blue and red indicate the mushy zone behind the molten pool. Tangents are drawn at the head and tail of the liquid molten pool, and the angle between their backward extensions is measured.

Under the same curvature of the LSSW path, the actual molten pool area undergoes continuous changes. Table 3 presents the radius and length of the molten pool measured along the inner and outer ring paths.

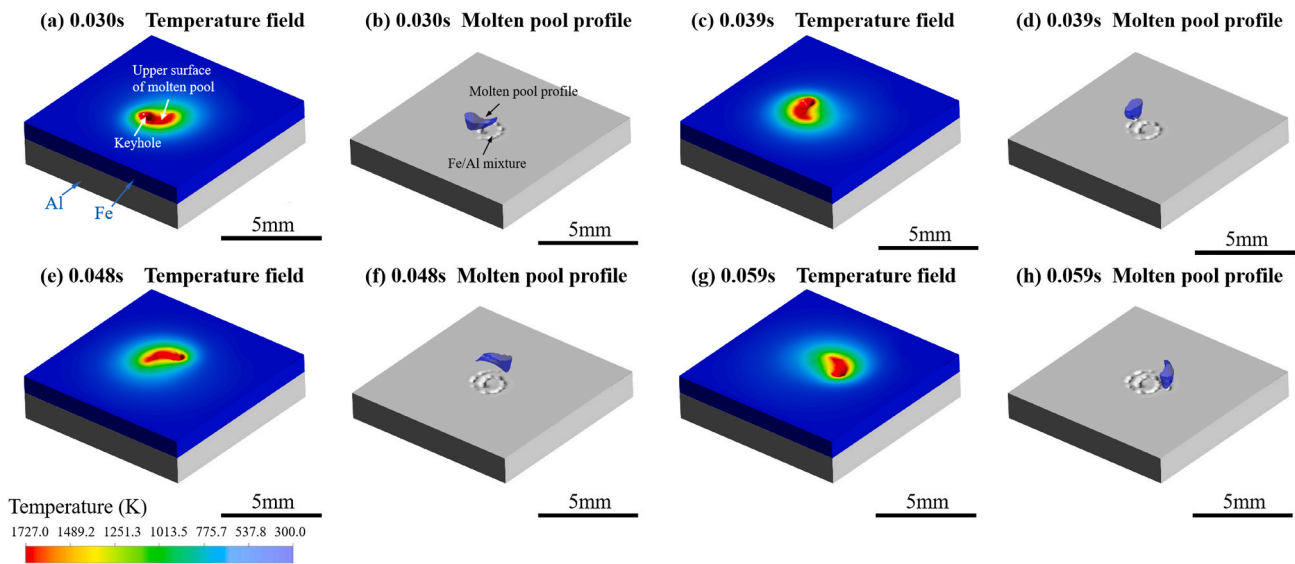


Fig. 5. Morphology of the upper surface and the Fe side of the molten pool of the inner ring weld at different times: (a) and (b) 0.030 s, (c) and (d) 0.039 s, (e) and (f) 0.048 s, (g) and (h) 0.059 s.

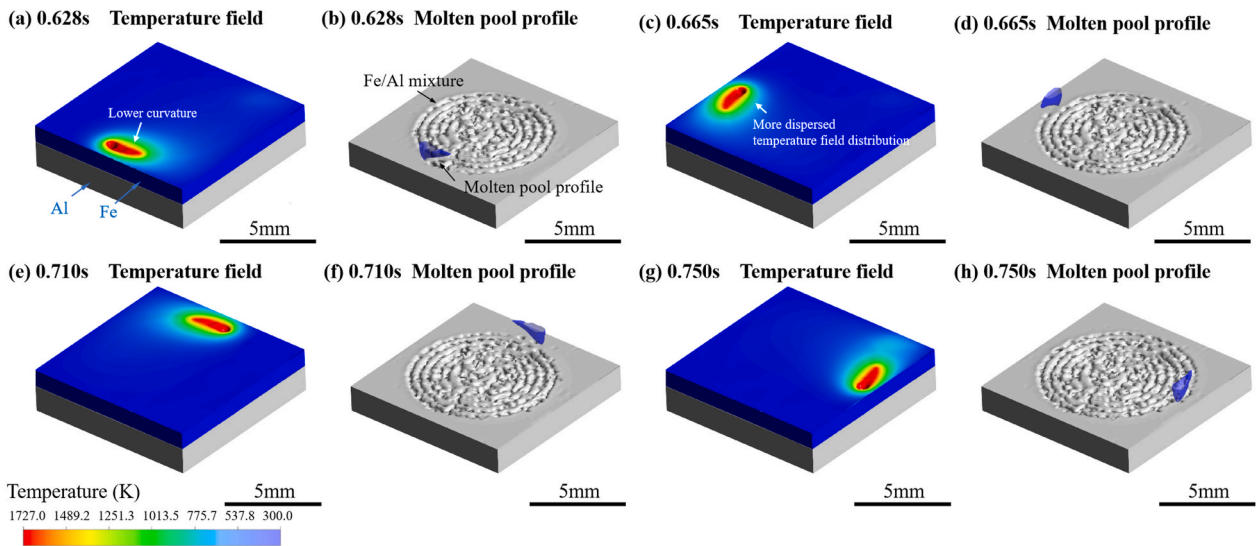


Fig. 6. Morphology of the upper surface and the Fe side of the molten pool of the outer ring weld at different times: (a) and (b) 0.628 s, (c) and (d) 0.665 s, (e) and (f) 0.710 s, (g) and (h) 0.750 s.

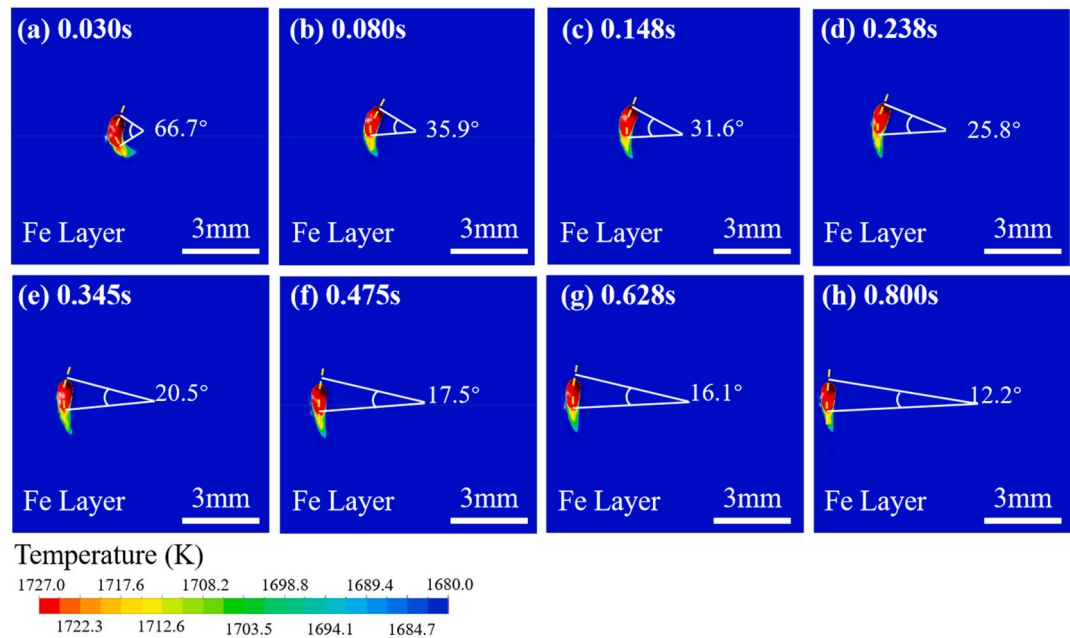


Fig. 7. Evolution of the upper surface shape of the Fe side molten pool under different welding orientations of the inner and outer rings: (a) 0.030 s, (b) 0.039 s, (c) 0.048 s, (d) 0.059 s, (e) 0.628 s, (f) 0.665 s, (g) 0.710 s, (h) 0.750 s.

Table 3
Curvature radius and length of the melt pool of the inner and outer rings.

Time (s)	Ring orientation	Radius (mm)	Molten pool length (mm)
0.030	2nd Left	1.09	1.27
0.039	2nd Right	1.29	1.26
0.048	2nd Right	1.49	1.04
0.059	2nd Down	1.39	1.43
0.628	8th Left	4.55	1.34
0.665	8th Top	3.57	1.12
0.710	8th Right	4.48	1.13
0.750	8th Down	3.57	1.16

The radius of the inner ring molten pool ranges from 1.09 mm to 1.49 mm, while its length varies between 1.04 mm and 1.43 mm. The corresponding average values are 1.32 mm and 1.25 mm, with variances of 0.29 and 0.28, respectively, indicating the degree of shape variation. For the outer ring, the molten pool radius and length fluctuate between 3.57 mm–4.55 mm and 1.12 mm–1.34 mm, respectively, yielding average values of 4.04 mm and 1.19 mm, and variances of 0.95 and 0.13. These data suggest that the molten pool lengths for both the inner and outer rings are generally consistent and exhibit only minor fluctuations.

The morphology of the molten pool is governed by the cooling rate, which is inversely proportional to the square of the temperature [19]. Consequently, the pool length is primarily determined by the laser power (P) and welding speed (v). Variations in heat input lead to

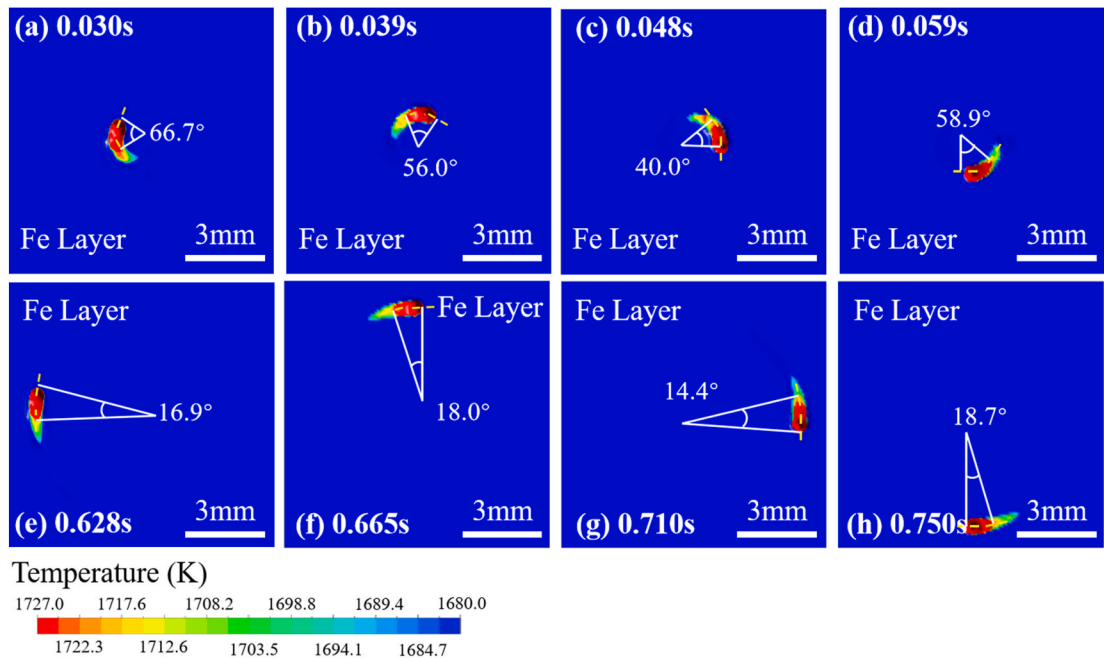


Fig. 8. Molten pool profile at different rings: (a) 0.030 s, (b) 0.080 s, (c) 0.148 s, (d) 0.238 s, (e) 0.345 s, (f) 0.475 s, (g) 0.628 s, (h) 0.800 s.

changes in temperature distribution and weld geometry [20]. When both parameters remain relatively stable during the welding process, the molten pool length shows limited variation. In contrast, the radius of the molten pool is largely influenced by the welding scanning path. The ratio of the inner to outer molten pool radii is 3.06, slightly lower than the ratio between the radii of the second (1 mm) and eighth (3.5 mm) circles of the spiral path.

However, the square difference of the molten pool radius increases significantly, and the front position of the molten pool is determined by the laser irradiation point. The stronger Marangoni force is generated at the tail of the molten pool under the larger temperature gradient, which leads to the increase of the heat flux fluctuation at the tail. Fig. 8 shows the morphological evolution of the molten pool top surface from the 2nd ring to the 9th ring at the same orientation. As the radius of the scanning path increases, the molten pool radius also increases accordingly.

As shown in Table 4, the increase in molten pool radius does not exhibit a one-to-one correspondence with the scanning path radius. The two values are only similar during inner-ring welding at 0.030 s. With the increase in welding rings, the molten pool radius (R_{mp}) is smaller than the scanning path radius (R_{os}), and the difference gradually enlarges. The molten pool is more curved than the scanning path, and the relationship between them can be described by the following linear expression:

$$R_{mp} = 0.74 \cdot R_{os} + 0.18 \quad (7)$$

Goodness of fit $R^2 = 0.98$ with high confidence level. Furthermore, Table 3 shows that the molten pool length fluctuates around 0.7 mm across different scanning radii and pass numbers, remaining roughly consistent.

Fig. 9 illustrates the temperature field distribution inside the molten pool along the 2nd and 8th circles. There exists an angle θ between the spiral scanning path and the direction of the maximum temperature gradient within the molten pool, and its magnitude is independent of the path radius. With the molten pool length remaining constant, the temperature gradient of the large radius molten pool in the tangential direction of the welding increases. It can also be observed in the figure that the length of the mushy zone between the solid phase line and the liquid phase line increases, resulting in a stronger Marangoni stress flowing to the tail of the molten pool along the tangential direction, thereby promoting an expansion in the flow radius of the molten metal. Therefore, a stable angle θ value makes a linear proportional relationship between the molten pool radius (R_{mp}) and the scanning path radius (R_{os}).

4.3. Interface characteristics of Al/ Fe mixed molten pool

Fig. 10 shows the temperature field distribution on the upper surface of the Al under the four welding directions of the inner ring (2nd ring) and the outer ring (8th ring) in the spiral scanning path. During the inner ring welding process from 0.030 s to 0.059 s, the temperature field on the Al layer is unstable, with the peak temperature at the center exceeding the liquidus temperature of the Al only at 0.039 s. The Al in the central area melts, and the liquid Al diffuses outward under the extrusion of the molten Fe. In this process, the two phases mix with each

other, forming an effective metallurgical bond. At other moments, the Al does not melt, and the Fe only reacts with the Al layer through atomic thermal diffusion.

However, due to the high energy density of the laser heat source, the heating and cooling rates are extremely fast. Under atmospheric pressure, effective metallurgical bonding is difficult to achieve relying solely on the mutual diffusion between Fe and Al. In the outer ring welding (Fig. 10(e)–(h)), the range of the heating zone is expanded, the number of moments when the heat source center exceeds the Al liquefaction temperature increased, and the forming continuity of the Al/Fe mixing zone along the welding path is enhanced.

4.4. Dynamic evolution of temperature/ flow field of Fe molten pool

The transverse cross-section temperature field and velocity field of the Fe side molten pool at 0.030 s and 0.628 s are extracted at 0.2 mm intervals. The schematic diagram of the extraction method is shown in Fig. 11. Fig. 11 (c, d) shows the temperature field and velocity field of the molten pool cross section at different distances from the keyhole center at 0.030 s and 0.628 s.

As displayed in Fig. 12 (a_T) and (a_V), the keyhole depth at this time was 0.7 mm, while the liquid molten pool depth was 0.9 mm, slightly smaller than the Fe layer thickness. The maximum flow velocity occurs on both sides of the keyhole wall, reaching up to 1.8 m/s. Although the molten pool depth was occasionally smaller than the Fe layer thickness, the temperature at the lower surface of the Fe layer remained higher than the liquidus temperature of the Al, ensuring continuous melting at the upper surface of the Al. Moreover, due to the small radius of the inner-circle welding path, the flow directions of the molten metal on the left and right sides of the keyhole different: the left side flows downward while the right side flows upward. Fig. 12 (b_T)–(f_T) and Fig. 12 (b_V)–(f_V) show the cross-sectional morphologies of the molten pool behind the keyhole. The keyhole radius was slightly larger than 0.2 mm. Beyond the direct laser irradiation zone, the liquid metal quickly backfilled and then gradually cooled down. The molten pool depth gradually decreased with increasing distance from the keyhole, and the flow velocity was similar among different cross-sections, all below 0.5 m/s.

Heat transfer within the molten pool is primarily achieved through thermal convection and conduction. The relative significance of convective heat transfer in the molten pool is evaluated using the Peclet number (Pe), which is defined as the ratio of thermal convection to thermal conduction [21].

$$Pe = \frac{\text{Heat}_{\text{convection}}}{\text{Heat}_{\text{conduction}}} = \frac{C_p \rho \mu L_R}{k} \quad (8)$$

where μ denotes the characteristic velocity (1.5 m/s selected around the keyhole and 0.5 m/s in the molten pool behind the keyhole). L_R represents the characteristic length. The cross-section of the molten pool is assumed to be rectangular, as illustrated in Fig. 12. The molten pool depth h corresponds to the height of the rectangle, while the average half-width $\bar{w}$ of the molten pool is taken as the length of the rectangle. Here, $\bar{w}$ is the average of half-widths measured at five different heights. The characteristic length can be expressed as:

$$L_R = \frac{4S}{C} = \frac{2\bar{w}h}{\bar{w} + h} \quad (9)$$

where C and S represent the wetted perimeter and cross-sectional area of the flow channel, respectively. Based on measurements, the $\bar{w}$ in the keyhole region is 0.25 mm, and the penetration depth is 0.9 mm, yielding L_R is 0.39 mm and the Peclet number is 41. In the region 0.8 mm away from the keyhole, $\bar{w}$ is 0.17 mm with a penetration depth of 0.3 mm, resulting in a Peclet number of 3. When the Peclet number is significantly greater than 1, heat transfer within the molten pool is dominated by convection. Conversely, conduction becomes the primary mechanism. Therefore, in most regions of the Fe-side molten pool along

Table 4

Curvature radius and length of the melt pool of between 2nd ~9th ring.

Time (s)	Scanning path radius (mm)	Molten pool radius (mm)	Molten pool length (mm)
0.030	0.5	0.63	0.73
0.080	1.0	0.99	0.62
0.148	1.5	1.27	0.70
0.238	2.0	1.51	0.68
0.345	2.5	2.03	0.73
0.475	3.0	2.37	0.72
0.628	3.5	2.61	0.73
0.800	4.0	3.38	0.72

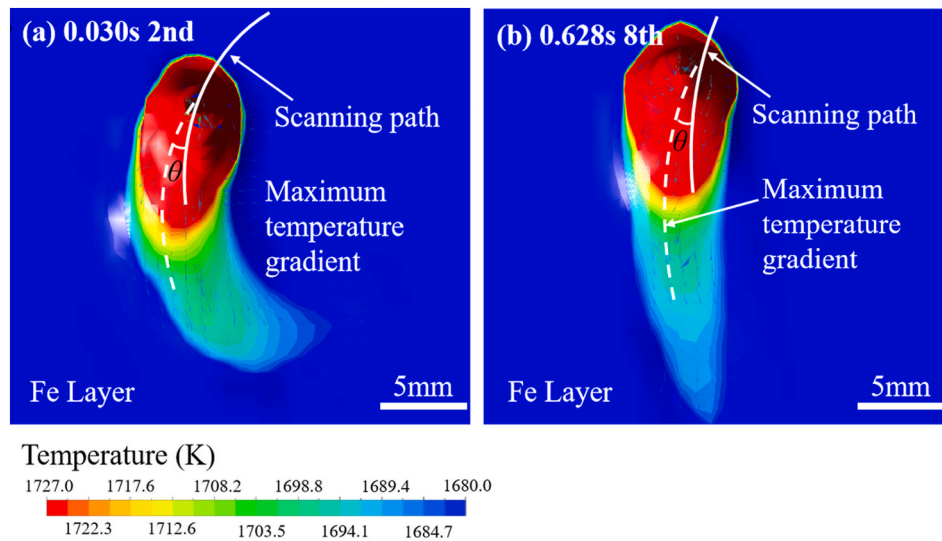


Fig. 9. Temperature field inside the molten pool and liquid metal flow field distribution under the (a) 2nd and (b) 8th spiral scanning path.

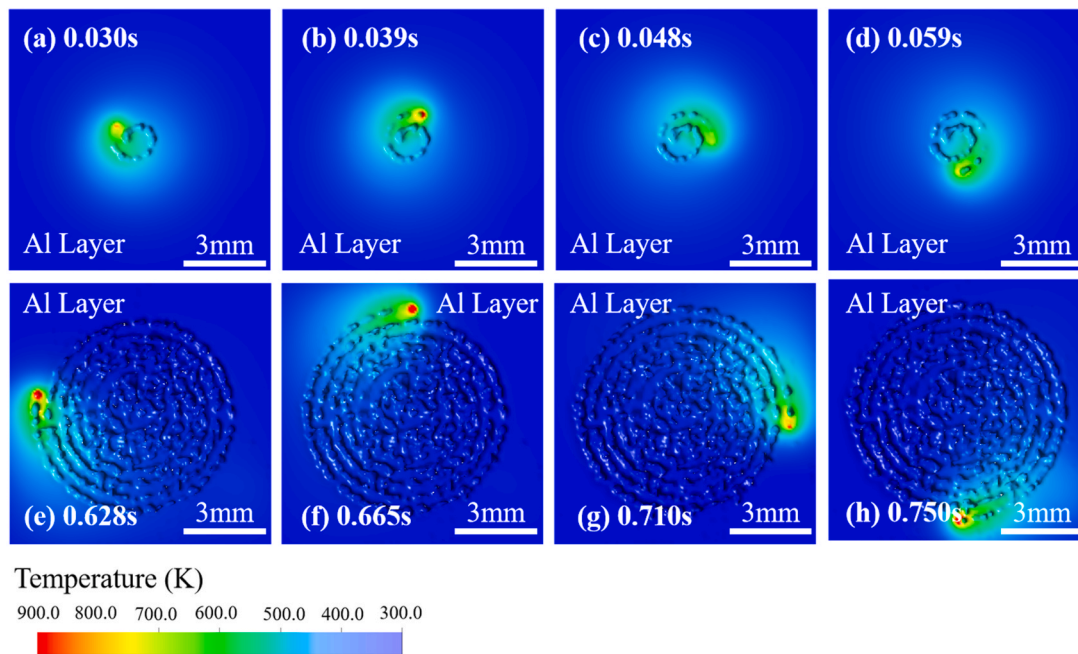


Fig. 10. Temperature field on the upper surface of the Al layer under different welding positions of the inner and outer rings: (a) 0.030 s, (b) 0.039 s, (c) 0.048 s, (d) 0.059 s, (e) 0.628 s, (f) 0.665 s, (g) 0.710 s, (h) 0.750 s.

the inner path of LSSW of Al/Fe dissimilar metals, the Peclet number ranges from 3 to 41, indicating that convective heat transfer is the dominant heat transfer mechanism.

The temperature and velocity field distributions of the spiral outer ring at 0.628 s are shown in Fig. 13. The change of the size of the molten pool in space is similar to that of the inner ring, characterized by a deeper front and a shallower rear. Comparing the flow velocity of the molten pool in each section, it can be seen that the flow velocity values of the inner and outer rings are similar, and the main difference is located around the keyhole.

In the outer region, the flow velocity near the keyhole decreases to approximately 0.6 m/s, and the flow direction on both sides is predominantly downward. It is distribution on both sides of the keyhole when the radius of the scanning path increases to approach a straight line, under which the liquid metal on the keyhole wall flows downward

driven by the evaporation recoil force. The Peclet number in most regions of the molten pool ranges between 2 and 20, indicating that convective heat transfer remains the dominant mechanism during outer-ring welding.

4.5. Dynamic evolution of temperature/ flow field of interface

Fig. 14 shows the phase distribution between the Al and the Fe in the transverse cross section. The two phases (Fe and Al) are intertwined within a small range near the interface, forming a periodically undulating wavy interface. Due to the limitations of the model employed in this study, which does not account for the thermal diffusion behavior of Al within the Fe molten pool, the simulated intermixing of the two metals results primarily from mutual convection between the rapidly flowing Fe liquid and the Al molten pool. In the Fe layer, the volume

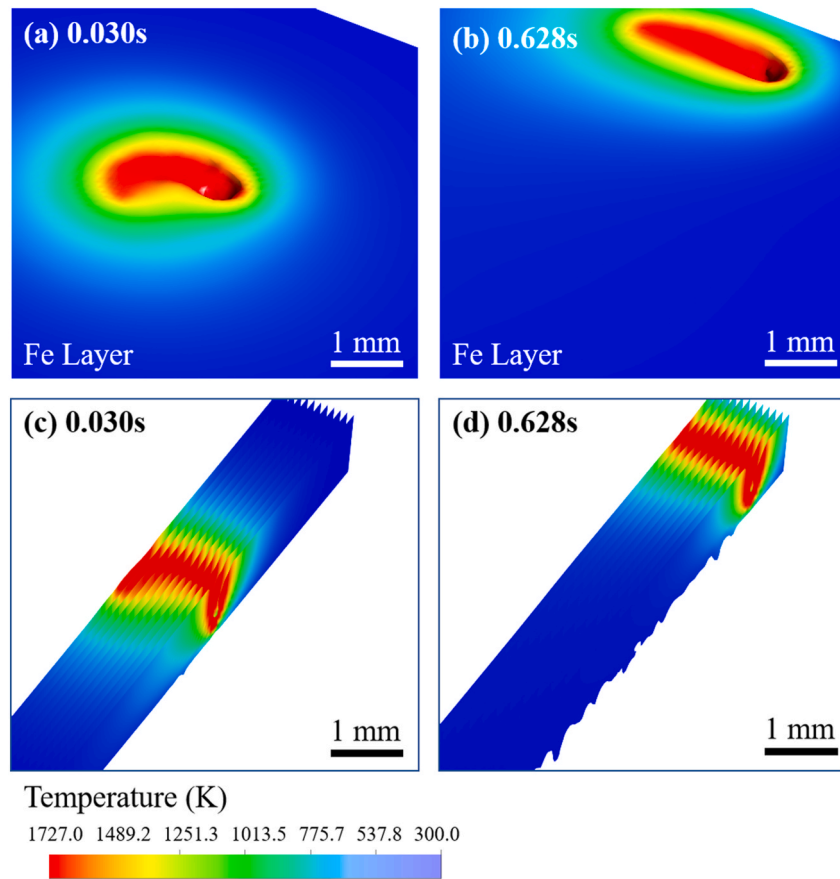


Fig. 11. Extraction method of temperature field and velocity field of molten pool cross-section: (a) and (c) 0.030 s, (b) and (d) 0.628 s.

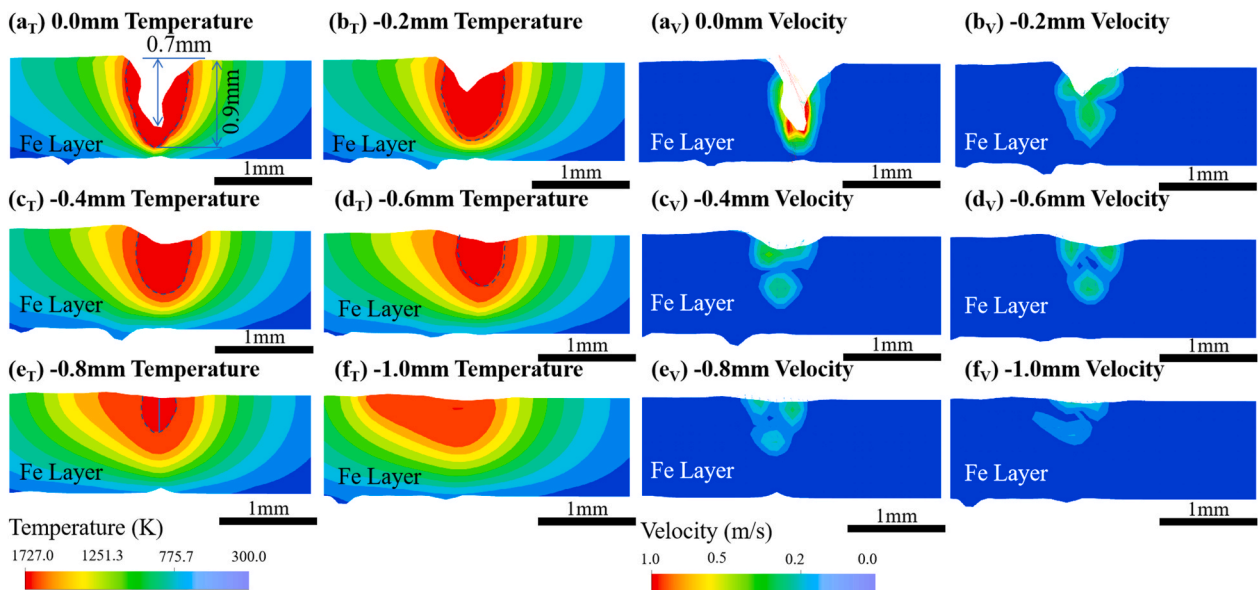


Fig. 12. Temperature and velocity field distribution of the 2nd circle molten pool cross section: (a) and (b) 0.0 mm, (c) and (d) –0.2 mm, (e) and (f) –0.4 mm, (g) and (h) –0.6 mm, (k) and (l) –1.0 mm.

fraction of Al is predominantly below 0.3, whereas the iron elements dissolved in the Al layer is distributed over a wider area, with a volume fraction exceeding 0.6. Among them, the area with the highest concentration is below the keyhole, which is the most intense area of molten pool flow.

Fig. 15 presents the temperature field distribution at the Al/Fe

interface. Fig. 15 (a) shows the morphology when the molten pool does not penetrate the Fe layer. Since the liquidus temperature of Fe is 1.9 times that of Al, the isotherm at 900 K (liquidus temperature of the Al) expands significantly more than the Fe molten pool. When the bottom of the Fe molten pool is 0.13 mm away from the Al/Fe interface, the 900 K isotherm on the Fe side coincides with the interface, defining this depth

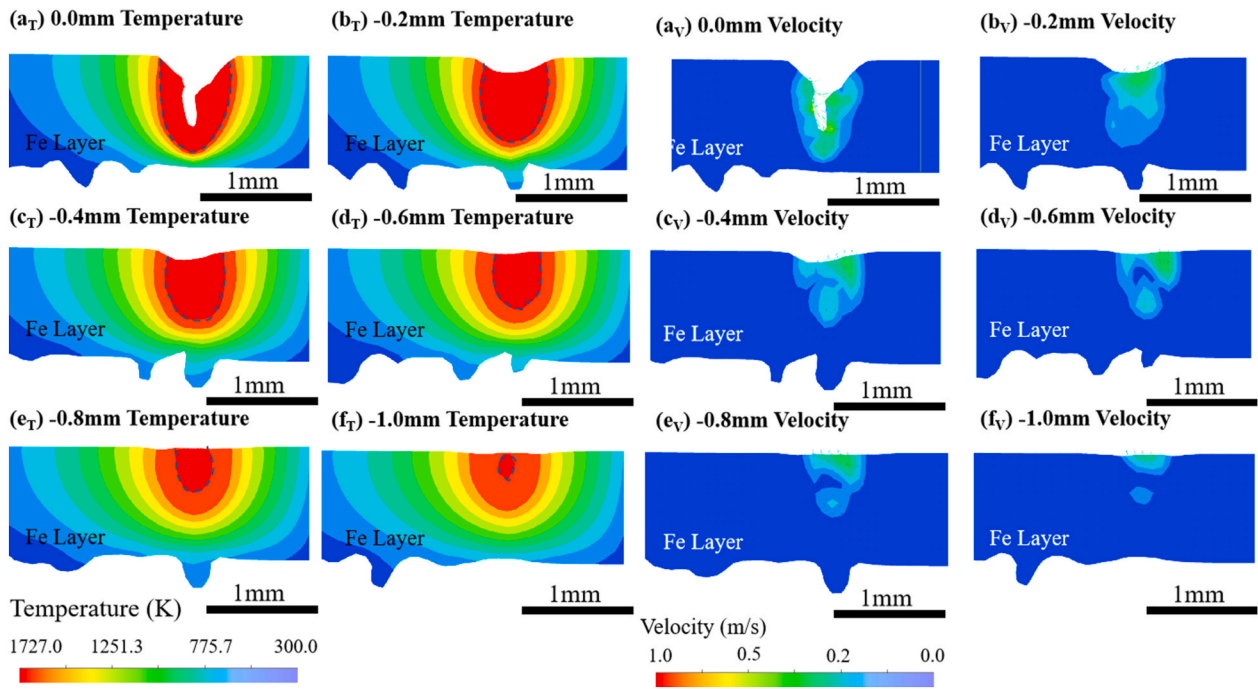


Fig. 13. Temperature and velocity field distribution of the 8th circle molten pool cross section: (a) and (b) 0.0 mm, (c) and (d) –0.2 mm, (e) and (f) –0.4 mm, (g) and (h) –0.6 mm, (k) and (l) –1.0 mm.

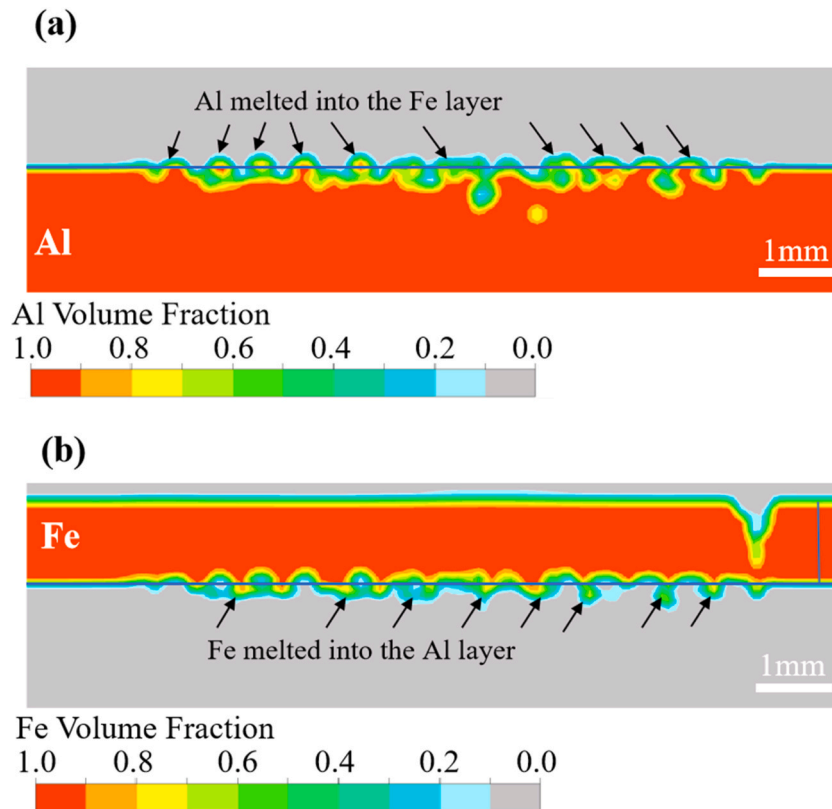


Fig. 14. Distribution of alloying elements at the Al/Fe interface: (a) Al, (b) Fe.

as the critical penetration threshold for melting the Al. Additionally, due to the thermal conductivity of Al is 4.5 times that of Fe, the width of the 900 K isotherm is significantly larger than the depth, showing a bowl shape with wide sides and shallow middle. With the increase in laser energy absorption below the keyhole, the keyhole depth increases, and

the recoil pressure from evaporation acts downward. It drives the liquid metal in the molten pool to flow downward, resulting in the molten pool morphology depicted in Fig. 15 (b). The surface layer of the Al melts and flows radially around under the impact of the liquid Fe to form a U-shaped liquid film with a thickness of only 0.05 mm. The liquid Al on

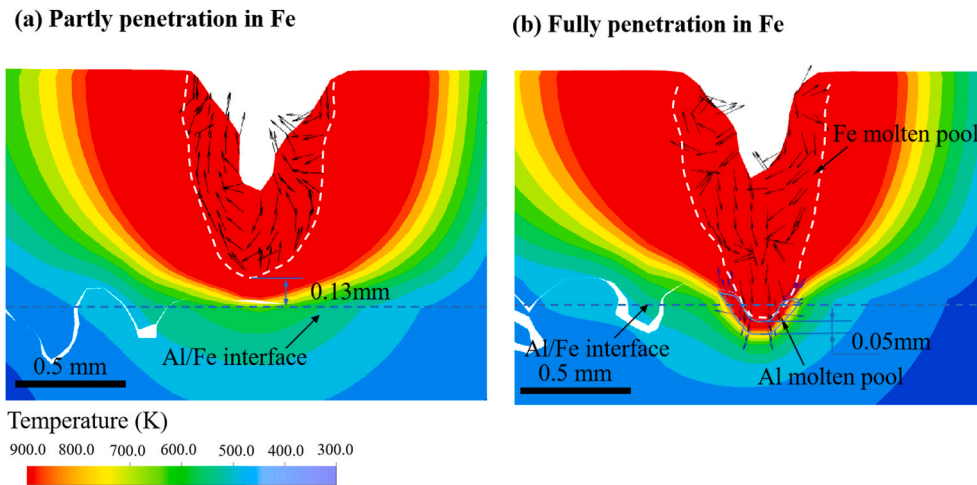


Fig. 15. Temperature field distribution and molten pool flow at interface under different penetration states (a) partially penetrated in Fe layer, (b) the fully penetrated in Fe layer.

both sides flows upward after being impacted by the liquid Fe, allowing the Al adjacent to the Fe molten pool to diffuse upward over a greater distance and form an Al-rich U-shaped solute band in the cross-section of the weld.

Fig. 16 shows the temperature history during the cooling process of the molten pool at $Z = 0.2$ mm in the 8th ring. Fig. 15 (a) illustrates the temperature field contour within the temperature range defined by the liquidus temperature of Al and room temperature. The temperature sampling line is positioned along the Y-axis between -0.0004 mm and -0.0016 mm, with a total of 1000 sampling points employed.

The extraction positions of Line 1 and Line 2 are shown in Fig. 16 (a). Fig. 16 (b) displays the temperature values at different points of the sampling line at various time instants. In the triangular region formed by the intersection of the Al liquidus, Line 1 and the temperature curve at 625.25 ms, the length of the right-angle edge in the x-axis direction represents the thickness of the liquid Al layer, while that along the y-axis corresponds to the temperature difference between the inner and outer surfaces of the liquid metal layer. The extremely small area of the triangle indicates that the liquid film of the Al layer is very thin, the temperature difference between the inner and outer ring is minimal, and its existence time is also extremely short. Furthermore, the region between Line 1 and Line 2 corresponds to the portion where molten Fe penetrates into the Al layer. Similarly, from the area enclosed by the Al

liquidus line, Line 2, and the 625.25 ms temperature curve, it can be inferred that, due to the substantial latent heat absorption during the melting phase transition of the Al and its high thermal conductivity, the molten Fe penetrating into the Al layer cools rapidly, with its solidification time being less than 0.25 ms. Finally, a comparison of the temperature values on both sides of Line 2 shows that the Fe at the interface solidifies prior to the Al.

5. Conclusions

In this study, a multi-physics numerical model is developed to elucidate the temperature evolution process and the molten pool flow mechanism in LSSW of Al/Fe lap joints. By combining the experimental and numerical results, the major conclusions of the work are summarized as follows:

- (1) Throughout the whole spiral welding path, the molten pool maintains an inverted triangular pyramidal morphology with curved edges, and its radius increases with the radius of the scanning path. The enhanced Marangoni stress in the outer ring leads to the molten pool radius smaller than the scanning path radius.

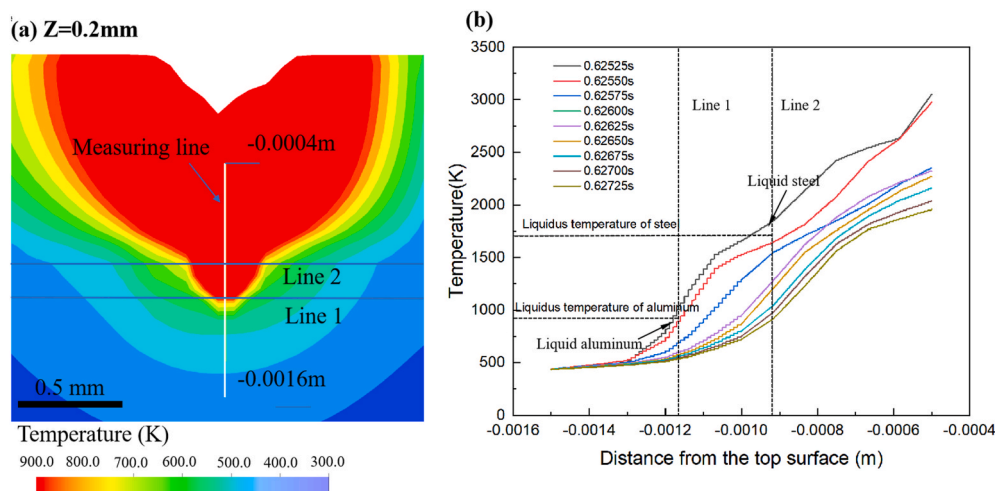


Fig. 16. Temperature history of the molten pool during cooling at $Z = 0.2$ mm in the 8th circle: (a) sampling position, (b) temperature change curve at different times at the sampling position.

- (2) The flow velocity in the inner pass is higher than that in the outer pass, with the maximum flow velocity occurring around the keyhole wall. The Peclet number in most regions of the molten pool ranges from 2 to 41, indicating that the convective heat transfer dominates the thermal exchange within the molten pool.
- (3) Driven by the evaporation recoil pressure, the Fe molten pool impacts the Al layer, partially melting and extruding the Al into a U-shaped liquid film, which is the primary reason for the formation of the Al-rich U-shaped solute band. Thermal history analysis indicates that the solidification time of the molten Fe penetrating the Al layer is less than 0.25 ms, and it solidifies prior to the Al film.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful to the supported by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (Grant Nos. JYB2025XDXM214)

References

- [1] Yang Y, Luo Z, Zhang Y, Su J. Dissimilar welding of aluminium to steel: a review. *J Manuf Process* 2024;110:376–97.
- [2] Walker L, Avedissian N, Vanderbilt C, Paolini D, Zhang W. Effect of Fe-Al intermetallics on fatigue properties of aluminum to steel dissimilar spot welds. *Int J Fatig* 2024;187:108434.
- [3] Yang J, Xie C, Zhang J, Qiao J. Design strategies for enhancing strength and toughness in ultrasonic welding of dissimilar metals: a review. *Mater Today Commun* 2025;42:111502.
- [4] Li A, Zeng Y, Fang P, Li Z, Guo Z, Yang Z, Wang Y, Liu Z, Cui J, Hong X. Effect of rare earth modification on the microstructure and mechanical properties of laser welding for DP1000 steel and 6061 aluminum alloy. *Mater Des* 2025;260:114998.
- [5] Yao Y, Jing L, Wang S, Li G, Cui J, Tang X, Jiang H. Mechanical properties and joining mechanisms of Al-Fe magnetic pulse welding by spot form for automotive application. *J Manuf Process* 2022;76:504–17.
- [6] Aizawa Y, Nishiwaki J, Harada Y, Muraishi S, Kumai S, J.J.o.M.P.. Experimental and numerical analysis of the formation behavior of intermediate layers at explosive welded Al/Fe joint interfaces 2016;24:100–6.
- [7] Shu F, Niu S, Zhu B, Wu L, Xia H, Chen B, Zhao J, Tan C. Influence of scan line spacing on nanosecond pulse laser welding of 6063 Al to steel thin sheets. *Opt Laser Technol* 2022;145:107497.
- [8] Sun J, Yan Q, Gao W, Huang J. Investigation of laser welding on butt joints of Al/steel dissimilar materials. *J Materials Design* 2015;83:120–8.
- [9] Xia H, Li L, Tan C, Yang J, Li H, Song W, Zhang K, Wang Q, Ma N. In situ SEM study on tensile fractured behavior of Al/steel laser welding-brazing interface. *Mater Des* 2022;224:111320.
- [10] Deng S, Yuan R, Tang X, Lu F. Migration behavior of IMC layer in twin-spot laser welding-brazing of aluminum to steel. *Mater Des* 2020;188:108489.
- [11] Xiong W, Zhang P, Yu Z, Yan H, Wu D, Lu Q, Tian Y. Weld zone porosity elimination process of galvanized steel zero-gap lap joints in remote laser spiral welding. *Mater Res Express* 2021;8:066502.
- [12] Li Q, Zhu B, Li H, Niu S, Wu L, Zeng Z, Xia H, Chen B, Tan C. Effect of spiral scan distance on the nanosecond-pulsed-laser lap joint of Al/Cu. *Opt Laser Technol* 2023;158:108896.
- [13] Wang X-N, Wang J-L, Sun Q, Tang X-X, Li X, Akira K, Huan P-C. Effects of nanosecond laser scanning paths on weld formation, microstructure and mechanical properties of dissimilar ultra-thin Al/Cu foil nanosecond laser welded joints. *J Mater Eng Perform* 2024;33:12043–53.
- [14] I R, S S, V L. Yb: YAG laser welding of dual phase steel to aluminium alloy. *J Mater Process Technol* 2018;262:411–21.
- [15] Pan J. Study on the behavior and mechanism of molten pool in LaserDeep fusion welding of steel/aluminum dissimilar metals. Hunan University; 2018.
- [16] Xie X, Zhou J, Long J. Numerical study on molten pool dynamics and solute distribution in laser deep penetration welding of steel and aluminum. *Opt Laser Technol* 2021;140:107085.
- [17] Jiménez-Xamán M, Hernández-Hernández M, Tariq R, Landa-Damas S, Rodríguez-Vázquez M, Aranda-Arizmendi A, Cruz-Alcantar P, J.F.i.M.E.. Numerical simulations and mathematical models in laser welding: a review based on physics and heat source models 2024;10:1325623.
- [18] Xu L, Tang X, Han S, Zhang R, Shao C, Cui H, J.J.o.M.P.T.. Study on root hump formation mechanism in full penetration laser welding with backing gas for aluminum alloy 2022;307:117674.
- [19] Hofmeister W, Griffith M. Solidification in direct metal deposition by LENS processing. *JOM* 2001;53:30–4.
- [20] Casalino G, Mortello M. A FEM model to study the fiber laser welding of Ti6Al4V thin sheets. *Int J Adv Manuf Technol* 2016;86:1339–46.
- [21] Fabbro R. Scaling laws for the laser welding process in keyhole mode. *J Mater Process Technol* 2019;264:346–51.